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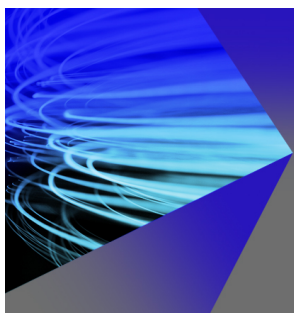
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Cite as: Phys. Fluids **36**, 027127 (2024); doi: [10.1063/5.0183475](https://doi.org/10.1063/5.0183475)

Submitted: 23 October 2023 · Accepted: 20 January 2024 ·

Published Online: 14 February 2024



View Online



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ABSTRACT

The packed beds are widely used in many industry sectors, such as energy, chemical engineering, and transportation. The prediction of fluid flow parameters is an important task for designing the packed bed units. In this study, the packed beds filled with spherical, cylindrical, and Raschig ring particles were experimentally and numerically investigated to understand the effect of pellet roughness on fluid flow parameters. The packed bed parameters were varied as follows: pellet roughness of 0.001–20 μm ; velocity of 1–10 m/s; D/d_p ratio of 2.9 and 4.8; packed bed length of 50–250 mm. It was established that the pressure drop increases significantly when the pellet roughness increases. Both numerical and experimental results showed a high deviation between our data and well-known empirical correlations. For this reason, a new correlation coefficient K for the Ergun equation taking into account pellet roughness was proposed. The correlation coefficient K provides a good correlation between the pressure drops calculated via the modified Ergun equation and the experimental/numerical data for a wide range of operational and design parameters.

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I. INTRODUCTION

Traditionally, flow through porous media or packed beds has been of interest in many areas of science and engineering, including oil reservoir development, food processing, chemical engineering (particularly filtration and catalytic reactors), and other scientific areas.^{1–3} Despite a deep understanding of the diverse processes occurring in packed beds, the number of research has remained unchanged over the years. This is due to the fact that various design and operating parameters, such as particle size and shape, L/D and D/d ratios, and many other factors, affected the flow characteristics in packed beds.^{4–6}

The main methods for the investigation of flow dynamics in packed beds are experiments and numerical simulations.^{7,8} Experiments provide sufficiently accurate results but it should be noted that real experiments are expensive and can take a lot of time to study the processes. At the same time, numerical modeling provides an

opportunity to carry out studies for a wide range of both design and operating parameters (velocity, pellet shape, type of fluid, etc.) without expensive experimental equipment and with much less time. Two main approaches of numerical investigation can be highlighted: particle-resolved when a flow over each particle of the packed bed is considered and pseudo-homogeneous model when the packed bed is modeled as a porous medium.^{9–11}

Nowadays, many equations for pseudo-homogeneous models have been proposed. The widely-used equations are the Ergun, Forchheimer, Blake–Kozeny, and Burke–Plummer equations.

The Ergun equation is described as the following:¹²

$$\frac{|\Delta P|}{L} = \frac{150\mu}{\phi^2 D_p^2} \frac{(1-\varepsilon)^2}{\varepsilon^3} v_\infty + \frac{1.75\rho}{\phi D_p} \frac{(1-\varepsilon)}{\varepsilon^3} v_\infty^2. \quad (1)$$

For laminar flow through a packed bed, the second term in the Ergun equation may be dropped, resulting in the Blake–Kozeny equation as follows:¹³

$$\frac{|\Delta P|}{L} = \frac{150\mu}{\phi^2 D_p^2} \frac{(1-\varepsilon)^2}{\varepsilon^3} v_\infty. \quad (2)$$

In these equations, v is a superficial gas velocity, μ is a viscosity, D_p is a mean particle diameter, L is a bed depth, ε is a void fraction, defined as the volume of voids divided by the volume of the packed bed region, and ϕ is a shape factor (sphericity).

For increasing filter velocities ($Re > 10$), inertial effects and turbulent friction forces are significant. In the Forchheimer equation,¹⁴ this fact is considered by adding an additional term to Darcy's law as follows:

$$-\Delta P = \frac{\mu}{k} V + \beta^2. \quad (3)$$

The Kozeny–Carman equation¹⁵ describes the flow through granular soils and packed beds by estimating the permeability of the porous medium from the porosity ε and the average particle diameter D_p

$$\frac{\Delta P}{L} = -\frac{180\mu}{\phi_s^2 D_p^2} \frac{(1-\varepsilon)^2}{\varepsilon^3} u_s. \quad (4)$$

The Burke–Plummer equation¹⁶ is applicable in flow regimes with the Reynolds number $Re > 1000$. Here, the pressure drop is proportional to the square of the velocity, L is the length of the packed bed, ΔP is the pressure drop, d_p is the effective (average) particle diameter in the porous medium, ε_p is the porosity, and v is the velocity magnitude

$$\frac{\Delta P}{L} = \frac{1.75}{d_p} \frac{(1-\varepsilon_p)}{\varepsilon_p^3} \rho v^2. \quad (5)$$

These manifold equations show that it is difficult to find a universal pseudo-homogeneous model for describing flows in a packed bed. The main reason for this fact is that the pressure drop depends on many parameters and take into account all of them into one equation is difficult. For example, the Ergun equation does not work well for some L/d and D/d , which is confirmed by the following works.^{17–19}

Empirical correlations are usually based on homogeneous assumptions with averaged characteristics and parameters. Therefore, they are often inapplicable to the packed beds with low D/d ratios when the main factor is the local phenomena in the packed bed^{20–23} and near the walls of both particles and the pipe wall.^{24–28} Local changes in the packed bed can lead to large deviations from empirical formulas in the predicted velocity profiles and, consequently, to the uneven pressure prod inside the packed bed. It is worth noting that forecasting the local void layer is also essential for predicting heat transfer and chemical reactions in packed beds, which is critical for data analysis. As was mentioned above, the Ergun equation for small D/d ratios gives a wrong pressure drop prediction and many equations were proposed for pressure drop for low D/d_p ratio papers.^{29,30}

An equation suggested by Reichelt²⁹ corrects Ergun's equation for wall effects

$$\frac{\Delta P}{L} = \frac{K_1 A_W^2 \mu_L V_L \varepsilon_S^2}{\phi^2 d_p^2 \varepsilon_L^3} + \frac{B_W \rho_L V_L^2 \varepsilon_S}{\phi_p d_p \varepsilon_L^3}, \quad (6)$$

where

$$A_w = 1 + \frac{2}{3(D/d_p)(\varepsilon_S)}, \quad (7)$$

$$B_w = \frac{1}{(k_1(d_p/D)^2 + k_2)^2}. \quad (8)$$

The coefficients K_1 , k_1 , and k_2 have been obtained by fitting the experimental data. For spheres, the authors proposed $K_1 = 154$, $k_1 = 1.5$, and $k_2 = 0.88$ and for cylinders $K_1 = 190$, $k_1 = 2$, and $k_2 = 0.77$.

For small diameter ratio and high Reynolds number, Wentz³⁰ proposed a correlation for pressure drop based on the Reynolds number that is given as

$$\frac{\Delta P}{L} = \left(\frac{\mu v_s}{d_p^2} \right) \left(\frac{(1-\varepsilon)^2}{\varepsilon^3} \right) \left(\frac{0.396 Re_m}{Re_m^{0.05} - 1.20} \right). \quad (9)$$

The packed bed is widely used in chemical engineering, in particular, methane reforming,^{31–35} water gas-shift reactions,^{36–38} etc. The catalyst pellets are used as particles in the packed beds. The distinguishing feature of catalyst pellets is their roughness. The roughness usually depends on the type of catalyst, material, fabrication, and support.^{39–41} In this case, the pore diameter determines the roughness, which can vary from a few micrometers to 1 mm.

The dependence of particle roughness on the pressure drop and other parameters in packed beds has often been investigated. These articles^{42–45} study how roughness affects turbulence, velocity, heat transfer, and fluid flow temperature. These papers show that roughness significantly depend on these parameters. In these papers,^{46,47} the

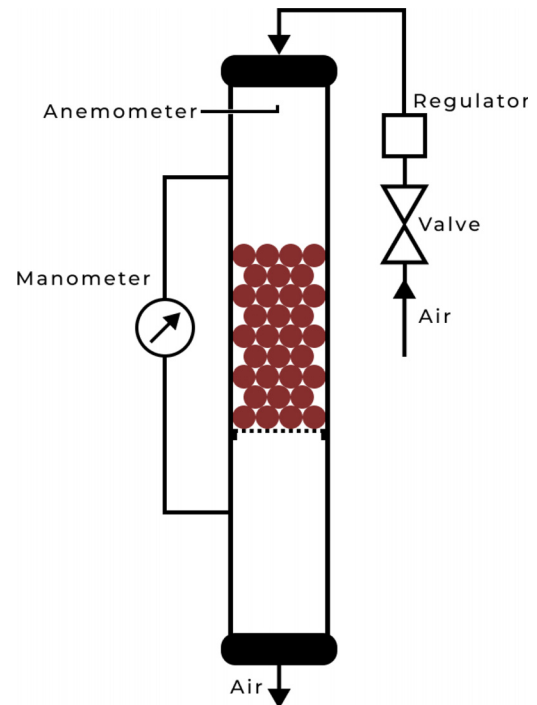


FIG. 1. Schematic diagram of the experimental setup.

dependence of roughness on different flow regimes was investigated. The papers propose an equation to describe the flow through packed beds of rough and smooth particles. The equation uses the effective surface area per unit volume of the packed bed. The authors reported that the roughness at small Reynolds numbers ($1.2 < Re < 63$) has a significant impact on the pressure drop in the packed bed.

Conducting our previous research on flow dynamics through packed beds filled with rough catalyst particles,^{7,48,49} we did not find a universal equation for pressure drop as a function of packed bed parameters and particle roughness for low D/d_p ratios. The present work evaluates the dependence of flow characteristics on the roughness of packed bed particles. Specifically, experiments and numerical simulations were conducted to obtain an equation for a pseudo-homogeneous model. The objective of this study was to experimentally and numerically determine the effect of surface roughness on the velocity – pressure drop relationship for gas flow through a packed bed and to obtain a semi-empirical equation for the pressure drop for low D/d_p ratios as a function of packed bed parameters and particle roughness.

II. METHODOLOGY

A. Experiments

The experimental setup shown in (Fig. 1) is used to investigate the pressure drop in a packed bed. The setup consists of pressure gauge

tubes, a cylindrical tube filled with pellets, an instrument panel displaying the flow pressure, and an Ostberg CK 200 axial fan. The inlet air velocity is controlled by the rotor speed of the fan and measured by a Testo Smart 405i anemometer. The pressure before and after the packed bed is measured by electronic manometers. Additionally, pellet roughness is measured using a profilometer, specifically the SURFTEST SV-3200 Production Series 178.

Different pellets were used to create packed beds, including spheres, cylinders, and Raschig rings. Pellets were either rough or rubbed (as shown in Fig. 2), with dimensions presented in Fig. 3. The measured pellet roughness is presented in Fig. 4. Packed beds are shown in Fig. 5, and their parameters are summarized in Table I.

Gas flow through the packed bed was tested at different air velocities, with measurements repeated at least twice for each velocity to evaluate the experiment accuracy. The pressure drop is determined as the average value from all repetitions. All experiments were performed at an ambient temperature of 25 °C.

Experiments were conducted for two types of pellets: rough pellets and rubbed pellets.

B. CFD-Modeling

The experimental and numerical investigation was performed at the same operating conditions and packed bed parameters. The

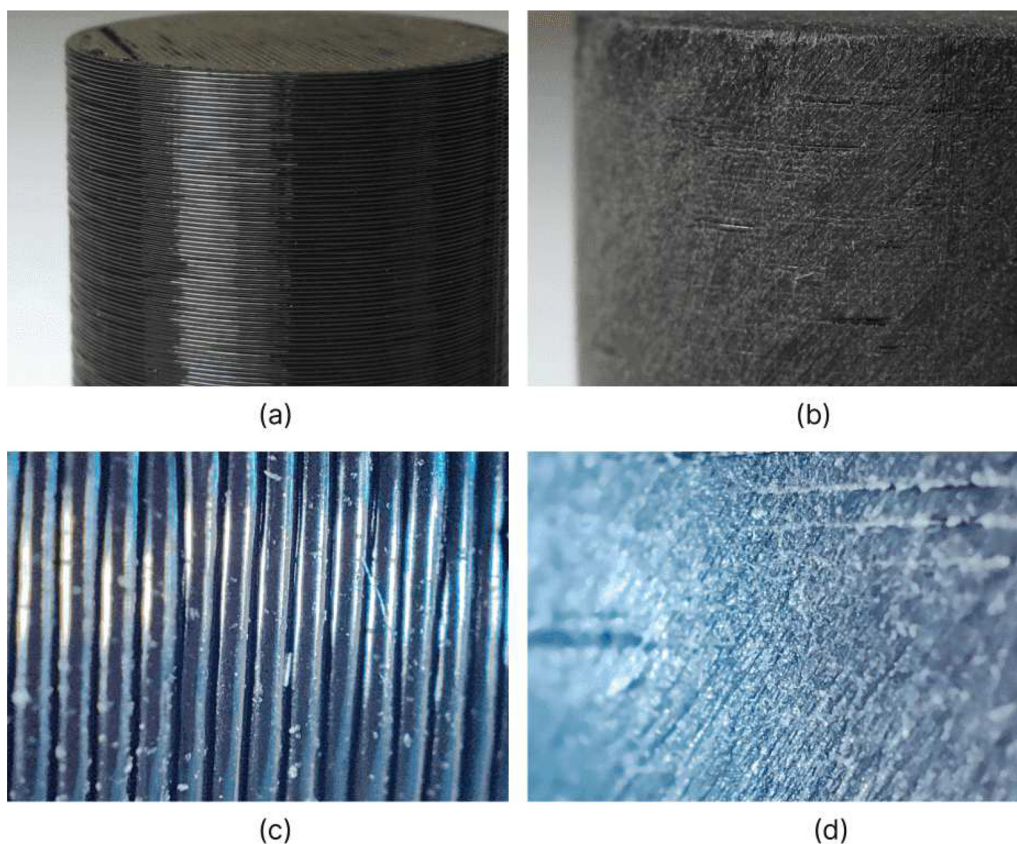


FIG. 2. The packed bed pellets: (a), (c) rough cylinder and (b), (d) rubbed cylinder.

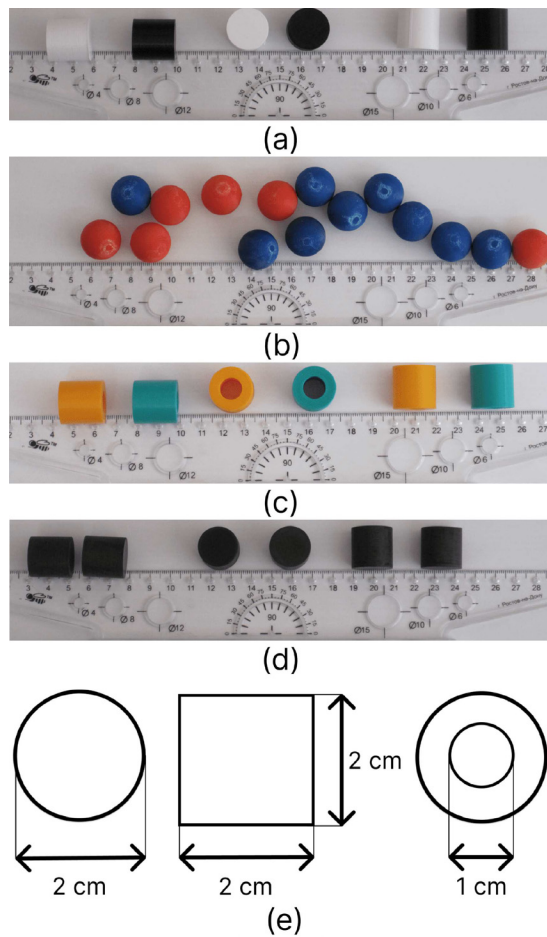


FIG. 3. The packed bed pellets: (a) cylinders, (b) spheres, (c) Raschig rings, (d) rubbed cylinders, and (e) pellet size.

computational fluid dynamics (CFD) model was realized using Ansys Fluent on Xeon E5-2640 with double 20-core 2.4 GHz processors with 48 GB random access memory (RAM). Convergence for each numerical calculation took approximately 500 iterations, with a computation time of about 40 min. Each CFD solution was conducted at least three times.

The three-dimensional stationary field of fluid flow through a packed bed is modeled by solving the Reynolds averaged Navier-Stokes equations (RANS), as well as the mass conservation equation. In this work, the commercial package Ansys Fluent was used. For the inlet velocity, the airflow in the tube with a packed bed is turbulent with the Reynolds number at the beginning of the pipe in the range from 61 538 to 307 692 for $D/d = 4.8$ and in the range from 37 180 to 353 205 for $D/d = 2.9$. The computational domain of the packed bed is presented in Figs. 6 and 7.

Figure 7 shows a few examples of packed beds investigated in this paper. Figure 7 shows a packed bed with the packing of spherical and cylindrical particles as well as Raschig rings. In Fig. 7 is the tube inner diameter (D), the height of the packed bed (H), the particle diameter (d), and the height of cylindrical particles (h).

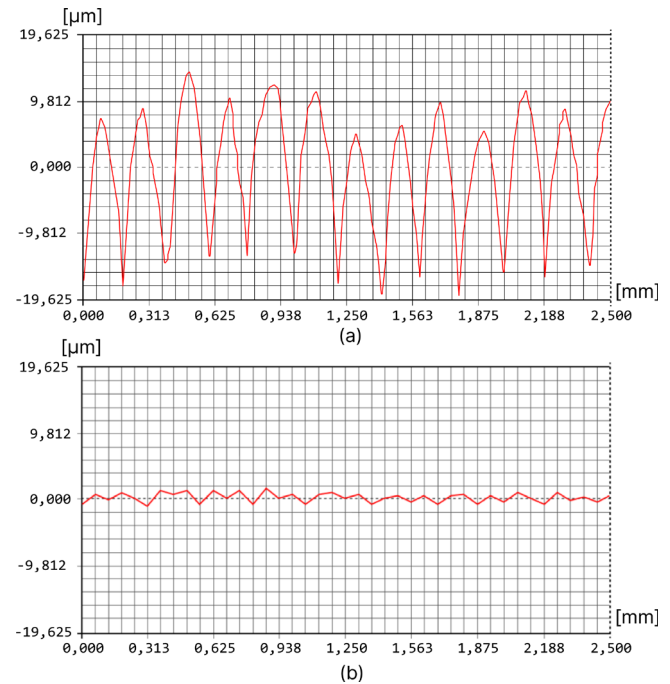


FIG. 4. Pellet roughness: (a) cylinders and (b) rubbed cylinders.

Python code was used for repeating the calculations of the pressure drop in each packed bed for various inlet velocities.⁵⁰ The $K-\omega$ model of the shear-stress transport (SST) group was chosen.

The turbulence model $k-\omega$ is more adapted to the calculation of problems with roughness.^{51,52} The turbulence model $k-\omega$ is more adapted to the calculation of problems with roughness.^{51,52} This study determined that the SST turbulence model with four equations gives the best result compared to the real experiment. However, nevertheless, the SST turbulence model requires more time and resources for the calculations. Consequently, the $k-\omega$ SST turbulence model was chosen for CFD modeling.

1. The packed bed generation

There are many methods to generate a packed bed. The discrete element method (DEM) is one of the most widely used. DEM is a numerical scheme that models particles' dynamic and static behavior based on contact mechanics. The soft particle model is used in DEM to simulate the behavior of particles in contact with points and friction sliders. The motion of each particle is tracked, and its interaction with other particles or boundaries is considered.

In this study, the packed bed was generated via Python code written by the authors of this paper. In our future papers, we will describe that code in detail. An example of a packed bed generated via Python code is shown in Fig. 8(a). The Python code provides an opportunity to develop the packed bed without DEM. It also allows the pellets to be used in various shapes and sizes because a mathematical equation describes them. The results of this work confirm that the code models the packed bed quite accurately because the number of pellets in the

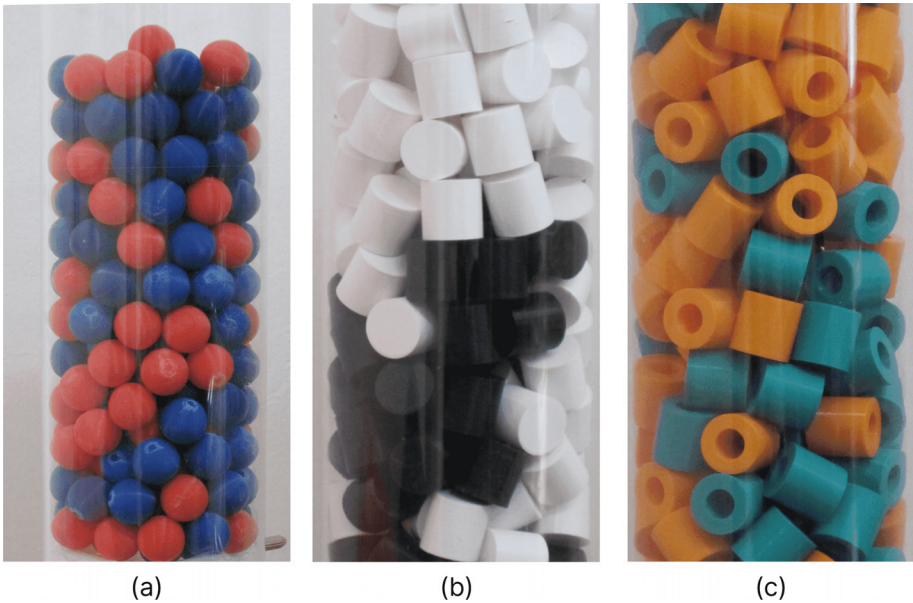


FIG. 5. The packed beds filled with (a) spheres, (b) cylinders, and (c) Raschig rings.

TABLE I. The packed bed parameters.

Parameters	$D/d_p = 4,8$	$D/d_p = 2,9$
Packed bed diameter (D), mm	95	58
Particle size (d_p), mm	20	20
Inlet and outlet diameters, mm	95	58
Number of particles	35–205	10–39
Fluid	Air	Air
Packed bed length, mm	50–250	50–250
Distance between the inlet and the particle bed starting point, mm	130	65
Distance between the outlet and the particle bed ending point, mm	410	130
Total length of the geometry, mm	1000	600

experiment and CFD are the same. It is worth noting that the Python code allows modeling the packed bed much faster than when using the Gilbert–Johnson–Keerthi algorithm (GJK) (the algorithm used in Blender) and DEM.

2. Mesh

A polyhedral unstructured mesh is generated using the Ansys Meshing and is presented in Fig. 8(b). A near-wall layer was added near pellets to calculate the pressure drop better. The inflation layer is also added near the tube wall in the fluid domain. The edge size of the elementary cells is the same for all computational domains. The meshes for all cases are generated with the same mesh settings. The total thickness of the inflation layer is adjusted to get $20 < y^+ < 300$. This parameter is a requirement for the $k-\omega$ SST turbulence model.^{51,52} A mesh independence study was performed for the fluid domain using various numbers of mesh elements. Table II shows the

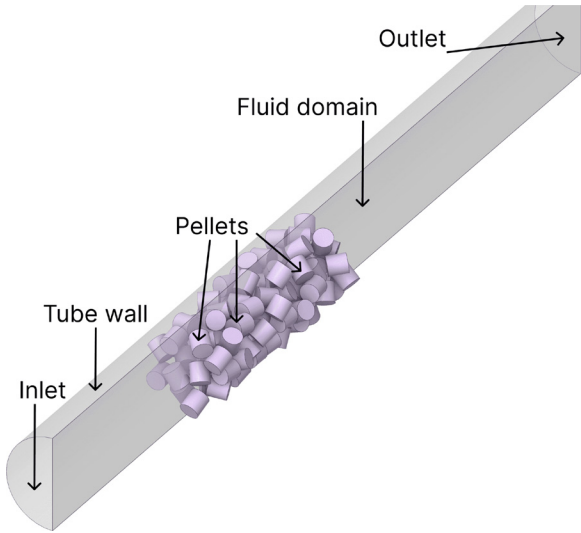


FIG. 6. A computational domain of a packed bed that is filled by pellets with a cylindrical shape.

effect of the number of elements on the pressure drop. The number of elements in the fluid domain was checked and was 6.51×10^6 .

3. Governing equation

A single-phase turbulent three-dimensional steady-state flow is modeled by solving the Reynolds-averaged Navier–Stokes equations. The following assumptions are made in CFD modeling:

- steady-state conditions;
- no energy flow;
- no work due to viscosity and pressure forces.

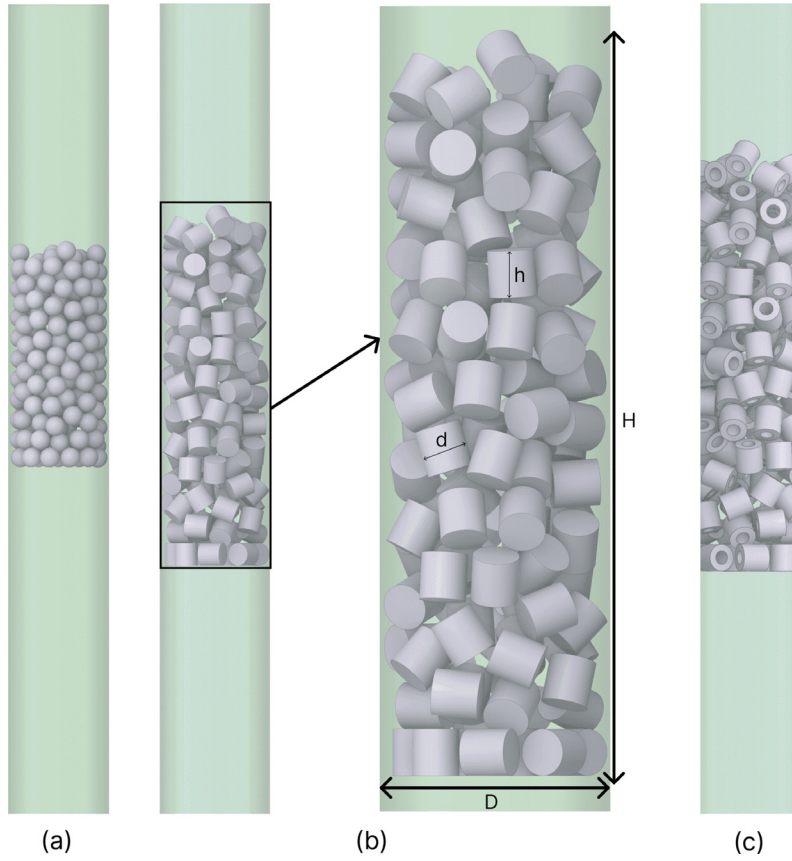


FIG. 7. The computational domains of the packed beds filled with (a) spheres, (b) cylinders, and (c) Raschig rings. H is the packed bed length, D is the pipe diameter, h is the cylinder height, and d is the cylinder diameter.

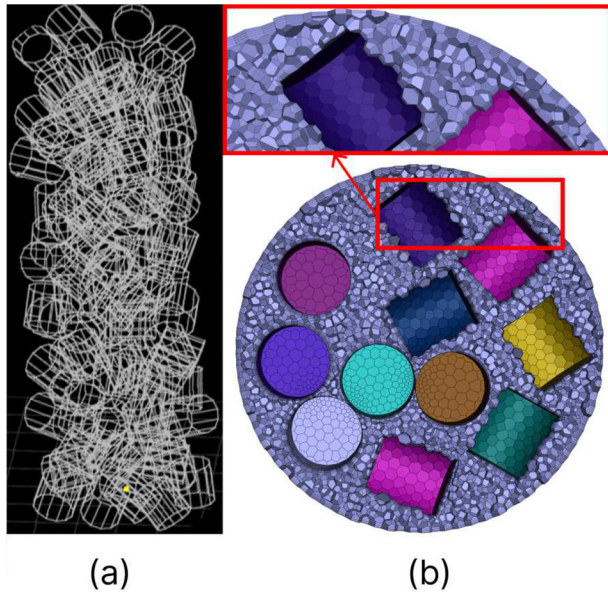


FIG. 8. (a) A fragment of the packed bed filled with cylinders generated via Python code and (b) mesh structure for the packed bed filled with cylinders.

TABLE II. Influence of mesh elements on the pressure drop for the cylindrical pellets for the case at $D/d = 2.9$.

Number of elements	Pressure drop [Pa]	Errors [%]
2.12×10^6	390	13.33
4.87×10^6	415	6.51
5.74×10^6	428	3.14
6.51×10^6	430	2.79
7.53×10^6	432	2.51

The mass conservation equation, or continuity equation, for steady-state conditions can be written as follows:

$$\nabla \times (\rho \vec{v}) = 0. \quad (10)$$

For the fluid region, the conservation of momentum is described by the following equations:

$$\nabla \times (\rho \vec{v} \vec{v}) = -\nabla p + \nabla \times (\vec{\tau}) + \rho \vec{g} + \vec{F}, \quad (11)$$

$$\vec{\tau} = \mu \left[(\nabla \vec{v} + \nabla \vec{v}^T) - \frac{2}{3} \nabla \times \vec{v} I \right]. \quad (12)$$

The mathematical description of the k - ω SST model can be expressed as follows:

$$\frac{\partial}{\partial t}(\rho\omega) + \frac{\partial}{\partial x_i}(\rho\omega u_i) = \frac{\partial}{\partial x_j} \left(\Gamma_\omega \frac{\partial \omega}{\partial x_j} \right) + G_\omega - Y_\omega + D_\omega + S_\omega, \quad (13)$$

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left(\Gamma_k \frac{\partial k}{\partial x_j} \right) + \tilde{G}_k - Y_k + S_k. \quad (14)$$

The equation of the effect of roughness in Ansys is

$$\frac{u_p u^*}{\tau_w / \rho} = \frac{1}{\kappa} \ln \left(E \frac{\rho u^* y_p}{\mu} \right) - \Delta B, \quad (15)$$

where

$$u^* = C_\mu^{1/4} k^{1/2} \quad (16)$$

and

$$\Delta B = \frac{1}{\kappa} \ln f_r. \quad (17)$$

In general, ΔB depends on the type (uniform sand, rivets, threads, ribs, wire mesh, etc.) and the size of the roughness. There is no universal roughness function that applies to all types of roughness. However, for sand-grain roughness and similar types of uniform roughness elements, ΔB is well correlated with the non-dimensional roughness height.

The non-dimensional roughness height can be expressed as

$$K_s^+ = \rho K_s u^* / \mu, \quad (18)$$

where

$$u^* = C_\mu^{1/4} k^{1/2}. \quad (19)$$

Experimental data analyses indicate that the roughness function is not a single function of K_s^+ . Instead, it takes different forms depending on the value of K_s^+ . Three distinct regimes have been observed given as follows:

- fully rough ($K_s^+ > 90$);
- hydrodynamically smooth ($K_s^+ \leq 2.25$);
- transitional ($2.25 < K_s^+ \leq 90$).

Based on the data presented, it can be observed that roughness effects are negligible in the hydrodynamically smooth regime. However, they become increasingly significant in the transitional regime and have a full impact in the fully rough regime. Ansys Fluent subdivides the entire roughness regime into three regimes and uses the formulas proposed by Cebeci and Bradshaw, based on Nikuradse's data, to calculate ΔB for each regime.

For the hydrodynamically smooth regime ($K_s^+ \leq 2.25$),

$$\Delta B = 0. \quad (20)$$

For the transitional regime ($2.25 < K_s^+ \leq 90$),

$$\Delta B = \frac{1}{\kappa} \ln \left[\frac{K_s^+ - 2.25}{87.75} + C_s K_s^+ \right] \times \sin \{ 0.4258 (\ln K_s^+ - 0.811) \}. \quad (21)$$

In the fully rough regime ($K_s^+ > 90$),

$$\Delta B = \frac{1}{\kappa} \ln (1 + C_s K_s^+). \quad (22)$$

In the solver, given the roughness parameters, $\Delta B(K_s^+)$ is evaluated using the corresponding formula. The modified law of the wall is then used to evaluate the shear stress at the wall and other wall functions for the mean temperature and turbulent quantities.

The packed bed porosity, ε , is defined as the fraction of the void volume within the packing height. For the packed beds with spherical particles, the packing porosity can be expressed as

$$\varepsilon = 1 - \frac{2Nd^3}{3D^2H}. \quad (23)$$

For the packed beds with cylindrical particles and Raschig rings, the packing porosity can be expressed as

$$\varepsilon = 1 - \frac{Nd^2h}{D^2H} \quad (24)$$

and

$$\varepsilon = 1 - \frac{N \times (d_1^2h - d_2^2h)}{D^2H}. \quad (25)$$

The porosity for the case $D/d = 2.9$ is shown in Table III.

The predicted pressure drops using CFD agree well with the experimental measurements, although the CFD predictions are generally lower than the measured results. The average error (5%) is within the desired range (10%–15%). This good agreement between predictions and measurements confirms the validity of the CFD model.

III. RESULTS

Figure 9 shows the typical CFD simulation results of a packed bed with cylindrical and spherical pellets, including streamlines on pellets and pressure contours in the packed bed. Each point on the pressure profile plot corresponds to one spatial position in the packing. The velocity magnitude is significantly smaller between particles than the free space velocity magnitude. In the present work, the velocity magnitude is the average velocity across the cross-section of the bed.

The experimentally measured pressure drops for the case of spherical shapes with a packed bed of 0.05 m at $D/d = 2.9$ case are plotted together with the corresponding CFD predictions and empirical correlations for direct comparison. They are shown in Fig. 10. No difference was found between CFD packing and experimental packing. The particle diameter in CFD decreased by 5% for spherical-type pellets and did not decrease for cylinders and Raschig rings.

Figure 10 confirms that empirical correlations are unreliable in predicting the pressure drop in a compacted bed with a low ratio of the pipe diameter to the particle diameter ($D/d < 5$), both with and without the presence of roughness. Moreover, Fig. 10 shows that the

TABLE III. The porosity of the packed bed for $D/d = 2.9$.

	Spheres	Cylinders	Raschig rings
D/d	2.9	2.9	2.9
ε	0.54–0.61	0.58–0.66	0.65–0.71

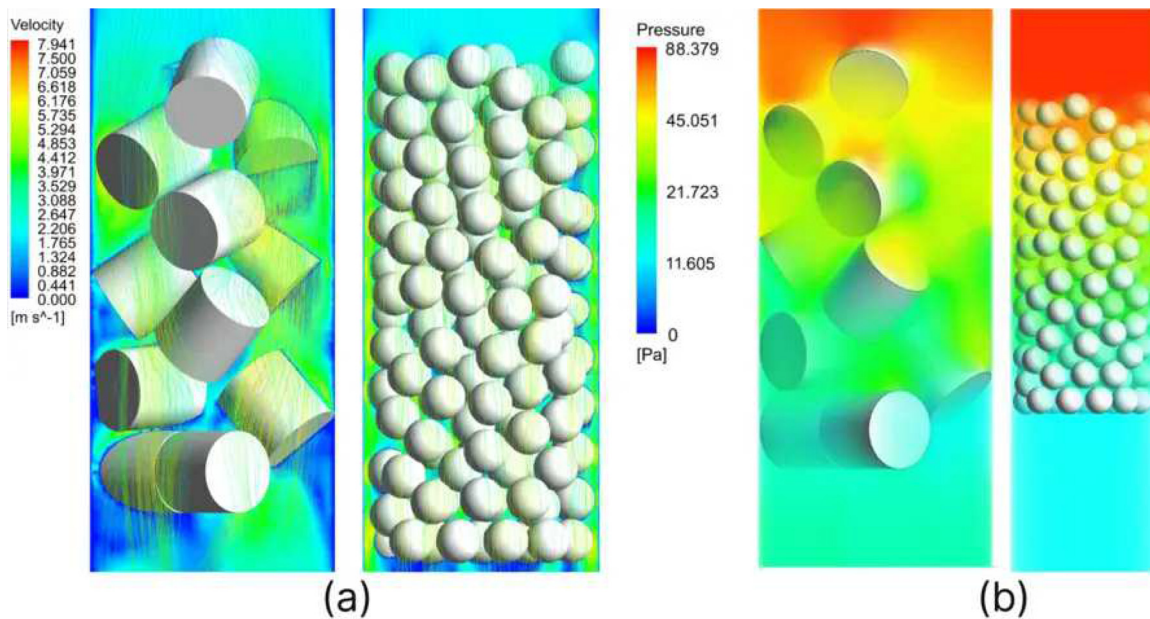


FIG. 9. The streamlines (a) and pressure contours (b).

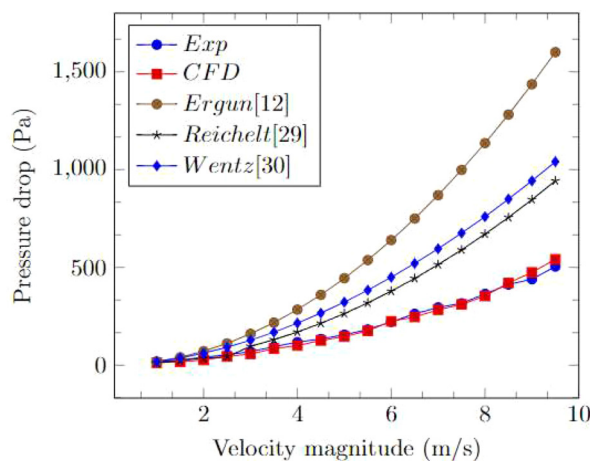


FIG. 10. Comparison of the numerical, experimental, and empirical data for the packed bed filled with spherical pellets: $L = 0.05$ m and $D/d = 2.9$.

Ergun correlation for predicting pressure drop in the packed bed is more unreliable than the correlations suggested by Reichelt²⁹ and Reger *et al.*³⁰ because these correlations are suggested especially for describing the pressure drop in the packed bed with a small ratio of D/d . This result once again confirms that the Ergun equation cannot be used to describe the pressure drop with small D/d . Their deviations are critical and protocontradictory and may significantly exceed the acceptable error limit.

The CFD results for pellets without roughness are plotted in Fig. 11 for comparison with the experimentally measured pressure drops. Figure 11 shows the results for the laboratory case with

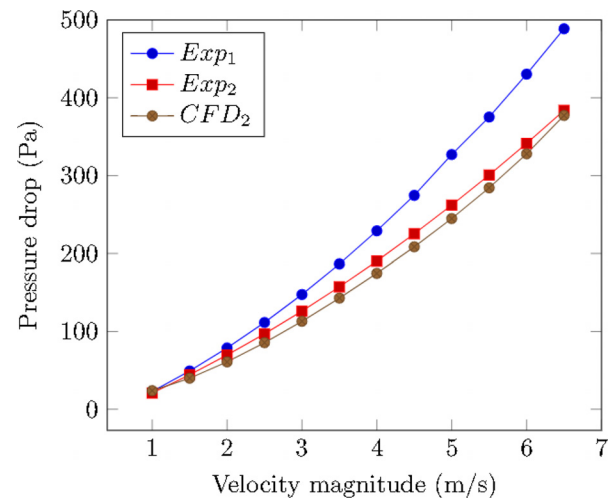


FIG. 11. The comparison of the experimental and numerical methods for cylindrical pellets with and without roughness for the case at $D/d = 2.9$. Exp₁—pellets with roughness were investigated and Exp₂—pellets without roughness were investigated.

cylindrical particle packing without roughness; the CFD calculation of this case was compared to the experiment with cylinders with roughness ($D/d = 2.9$). The CFD-predicted pressure drops agree with the experimental measurements, although the CFD predictions tend to be lower than the measured results. The average error (7%) is within the desired range (10%–15%). The deviation for each of their CFD experiments for the cylindrical pellets is shown in Fig. 12. This good agreement between predictions and measurements confirms the validity of

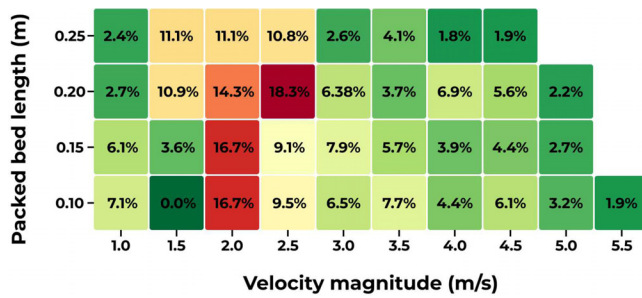


FIG. 12. The deviations between the experimental and numerical results for cylindrical pellets without roughness for all cases at $D/d = 2.9$.

the CFD model. Since the above results confirmed the validity of the CFD model for a particular layer, numerical simulations were performed for all layers on which the physical experiment was performed. The results of these numerical experiments are shown in Fig. 13.

Figure 13(a) shows that the effect of roughness does not have a relatively significant effect on the cylindrical type of pellets (10%–15%); at the same time a substantial difference between the values of spherical pellets and Raschig rings can be seen (50%). The significant difference in these values is most likely due to the 7% reduction of the spheres. Also, the large difference between the values of Raschig rings is probably due to the surface area of the pellet streamline. The Raschig rings have a higher surface area than cylinders or spheres, or a feature of the CFD modeling since the roughness was also set from the inside.

Figure 13(b) shows that the effect of roughness does not have a relatively large effect on the cylindrical type of pellets and Raschig rings, but at the same time, a strong difference between the values of spherical pellets can be seen. The large difference between these values is also most likely caused due to the 5% reduction of spheres in CFD. Obviously, the pressure drop line for spherical pellet particles and the lines for cylinders and Raschig rings should be higher. When processing the entire database, it was found that roughness consistently affects pressure drop by an average of 15%.

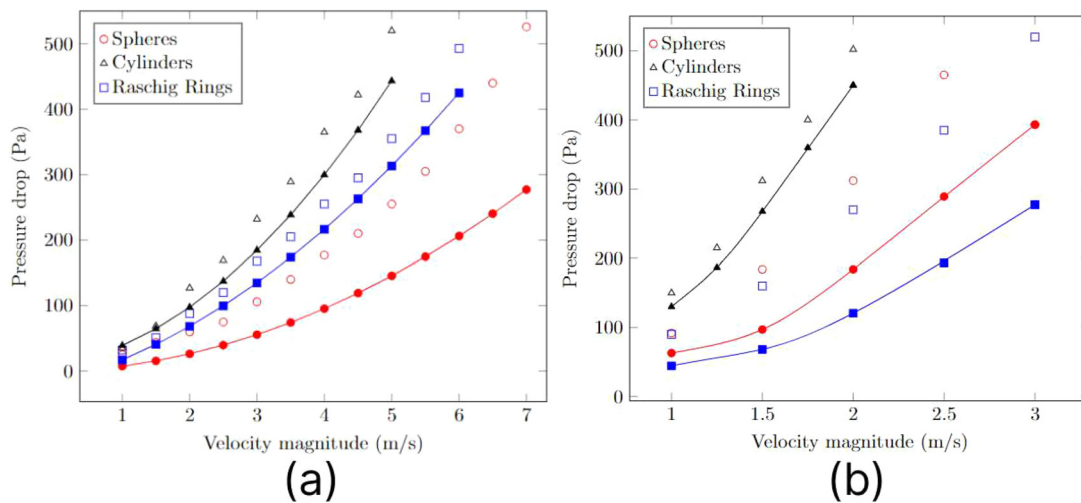


FIG. 13. Comparison of the experimental and numerical results for all types of pellets with and without roughness: (a) $D/d = 2.9$ and (b) $D/d = 4.8$. An empty marker is the experimental results, a filled mark on the line is the numerical results.

Based on all the results described above and since empirical correlations are unreliable, a coefficient K was derived for the Ergun formula for small diameters, which was determined empirically and algebraically. Then, the Ergun equation with coefficient K will be described as

$$\frac{|\Delta P|}{L} = K \times \left[\frac{150\mu(1-\varepsilon)^2}{\phi^2 D_p^2 \varepsilon^3} v_\infty + \frac{1.75\rho(1-\varepsilon)}{\phi D_p \varepsilon^3} v_\infty^2 \right], \quad (26)$$

where K is the correlation coefficient to consider the effect of the pellet roughness on the pressure drop in the packed bed. This correlation coefficient can be calculated as follows:

$$K = v^{0.015r-0.57}, \quad (27)$$

where r is the average roughness of pellets, and 0.57 and 0.015 are constants determined using the experimental and numerical data; these coefficients may have a deviation of 0.002, as this is the error of the anemometer in measuring the flow velocity in the pipe (confidence coefficient of 0.95), and v is the superficial gas velocity.

The coefficient K was created for the Ergun equation because this equation is the most popular for describing the pressure drop in the packed bed. Also, the Ergun equation can be acceptable for other liquids and gases because they are defined through viscosity and density. Therefore, this coefficient K is acceptable for describing the effect of roughness on the pressure drop for other liquids and gases.

The results with this coefficient are shown in Fig. 14. It can be seen that with the coefficient K , the empirical Ergun correlation gives qualitative results that correlate with the experimental and numerical data, and the error (7%) is within acceptable limits (10%–15%).

IV. CONCLUSION

The effect of pellet roughness on the pressure drop for flow through a packed bed was investigated both experimentally and numerically for a wide range of operational and design parameters. It was observed that particle roughness has a significant effect on

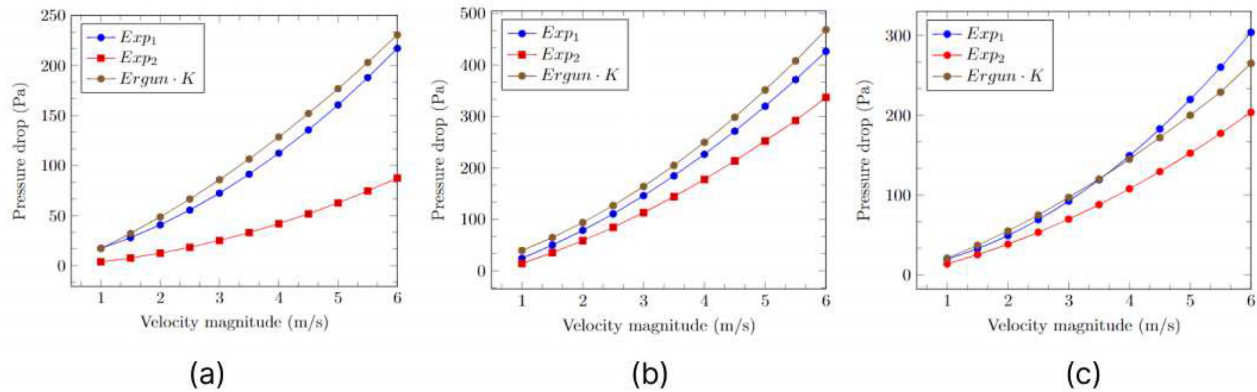


FIG. 14. The error of the results of the empirical Ergun formula with K coefficients and experimental results for (a) spherical pellets, (b) cylindrical pellets, and (c) Raschig rings without roughness for the case of $D/d = 2.9$. Exp_1 —pellets with roughness were investigated and Exp_2 —pellets without roughness were investigated.

pressure drop. For all investigated cases, a characteristic quadratic dependence of the pressure drop on velocity [$\Delta P = f(v^2)$] was obtained. The experimental and numerical results show that particle roughness can significantly (up to 17%) affect the pressure drop in packed beds. In this paper, a new equation for the pressure drop as a function of operational and design parameters, including particle roughness, was proposed for low D/d_p ratios.

A correlation coefficient $K = v^{0.015r-0.57}$ for the Ergun equation was derived to take into account particle roughness. The coefficient K was introduced based on experimental and numerical results to compensate for the effect of deviation of values obtained by the well-known Ergun equation. The results obtained via the new equation show a good correlation between experimental and numerical data with an average error of up to 7%. For the packed bed length of $L/D > 2$ and a velocity of 6–10 m/s, the relative error is below 2%. The tests were performed for a wide range of practically interesting cases: pellet roughness of 0.001–20 μm ; velocity of 1–10 m/s; D/d_p ratio of 2.9–4.8. Based on the obtained results, it can be concluded that for those operational and design parameters, our equation for correlation coefficient can be used for the accurate prediction of the pressure drop in packed beds.

ACKNOWLEDGMENTS

This work is supported by the Russian Science Foundation under Grant No. 23-79-10044.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Viacheslav Papkov: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (supporting); Validation (equal); Visualization (equal); Writing—original draft (lead); Writing—review & editing (equal). **Nikita Shadymov:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (lead); Validation (equal);

Visualization (equal). **Dmitry Pashchenko:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing—review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

NOMENCLATURE

C_s	Roughness constant (μm)
D_{ω}	Cross-diffusion term
d_1	External diameter of Raschig rings (mm)
d_2	Internal diameter of Raschig rings (mm)
D	Packed bed diameter (mm)
D_p	Particle diameter (mm)
d	Particle size diameter (mm)
$\vec{F}$	External body forces (N/m^3)
f_r	Roughness function that quantifies the shift of the intercept due to roughness effects
G_{ω}	Generation of the specific dissipation rate
G_k	Generation of the turbulence kinetic energy
H	Height of the packed bed (mm)
h	Particle height (mm)
I	Unit tensor
K	Correlation coefficient ($\text{m/s}^{\mu\text{m}}$)
K_s^+	Non-dimensional roughness height
K_s	Physical roughness height (μm)
k	Turbulence kinetic energy (m^2/s^2)
L	Packed bed length (mm)
N	Total number of particles in the packing
ΔP	Pressure drop (Pa)
p	Static pressure (Pa)
r	Average roughness of pellets (μm)
Re	Reynolds number
S_k	User-defined source term
S_{ω}	User-defined source term

V	Volume of the particle (m^3)
Y_ω	Dissipation of the specific dissipation rate due to turbulence
Y_k	Dissipation of the turbulence kinetic energy due to turbulence
ρ	Density (kg/m^3)
μ	Dynamic viscosity ($\text{Pa} \cdot \text{s}$)
μ_t	Eddy viscosity ($\text{Pa} \cdot \text{s}$)
Γ_ω	Effective diffusivity of the specific dissipation rate
Γ_k	Effective diffusivity of the turbulence kinetic energy
$\rho \vec{g}$	Gravitational body force [$(\text{kg} \cdot \text{m})/\text{s}^2$]
ε	Porosity
ω	Specific dissipation rate ($1/\text{s}$)
ϕ	Sphericity
$\bar{\tau}$	Stress tensor
v	Superficial velocity (m/s)

REFERENCES

- ¹Y. Feng, B. Gong, H. Cheng, L. Wang, and X. Wang, "Effects of fixed wall and pebble size ratio on packing properties and contact force distribution in binary-sized pebble mixed beds at the maximum packing efficiency state," *Powder Technol.* **390**, 504–520 (2021).
- ²S. Singh and J. Rautela, "Numerical study of packed bed thermal energy storage with natural sandstone rock as a filler material," *AIP Conf. Proc.* **2863**, 020008 (2023).
- ³T. M. Przybycien, N. S. Pujar, and L. M. Steele, "Alternative bioseparation operations: Life beyond packed-bed chromatography," *Curr. Opin. Biotechnol.* **15**, 469–478 (2004).
- ⁴Z. Guo, Z. Sun, N. Zhang, and M. Ding, "Influence of confining wall on pressure drop and particle-to-fluid heat transfer in packed beds with small D/d ratios under high Reynolds number," *Chem. Eng. Sci.* **209**, 115200 (2019).
- ⁵K. Allen, T. Von Backström, and D. Kröger, "Packed bed pressure drop dependence on particle shape, size distribution, packing arrangement and roughness," *Powder Technol.* **246**, 590–600 (2013).
- ⁶B. Einfeld and K. Schnitzlein, "The influence of confining walls on the pressure drop in packed beds," *Chem. Eng. Sci.* **56**, 4321–4329 (2001).
- ⁷D. Pashchenko, I. Karpilov, and R. Mustafin, "Numerical calculation with experimental validation of pressure drop in a fixed-bed reactor filled with the porous elements," *AIChE J.* **66**, e16937 (2020).
- ⁸R. K. Reddy and J. B. Joshi, "CFD modeling of pressure drop and drag coefficient in fixed beds: Wall effects," *Particuology* **8**, 37–43 (2010).
- ⁹N. Jurtz, M. Kraume, and G. D. Wehinger, "Advances in fixed-bed reactor modeling using particle-resolved computational fluid dynamics (CFD)," *Rev. Chem. Eng.* **35**, 139–190 (2019).
- ¹⁰B. Niu, M. Ahmed, T. Wen, G. Xu, and X. Liu, "Particle-resolved CFD modeling of the flow and heat transfer characteristics of the fluid in a fixed bed," *Powder Technol.* **421**, 118445 (2023).
- ¹¹Y. Shi, H. Chen, W. Chen, G. Ye, J. Qu, J. Li, X. Zhou, and X. Duan, "Effects of particle shape and packing style on ethylene oxidation reaction using particle-resolved CFD simulation," *Particuology* **82**, 87–97 (2023).
- ¹²S. Ergun, "Fluid flow through packed columns," *Chem. Eng. Prog.* **48**, 89 (1952).
- ¹³M. J. MacDonald, C.-F. Chu, P. P. Guilloit, and K. M. Ng, "A generalized Blake-Kozeny equation for multisized spherical particles," *AIChE J.* **37**, 1583–1588 (1991).
- ¹⁴S. Whitaker, "The forchheimer equation: A theoretical development," *Transp. Porous Media.* **25**, 27–61 (1996).
- ¹⁵R. P. Chapuis and M. Aubertin, "On the use of the kozeny carman equation to predict the hydraulic conductivity of soils," *Can. Geotech. J.* **40**, 616–628 (2003).
- ¹⁶S. P. Burke and W. Plummer, "Gas flow through packed columns¹," *Ind. Eng. Chem.* **20**, 1196–1200 (1928).
- ¹⁷A. Montillet, E. Akkari, and J. Comiti, "About a correlating equation for predicting pressure drops through packed beds of spheres in a large range of Reynolds numbers," *Chem. Eng. Process.* **46**, 329–333 (2007).
- ¹⁸D. Nemec and J. Levec, "Flow through packed bed reactors: 2. Two-phase concurrent downflow," *Chem. Eng. Sci.* **60**, 6958–6970 (2005).
- ¹⁹S. Zhang, Z. Wen, X. Liu, Y. Xing, and H. Zhang, "Gas flow characteristics through irregular particle bed with the vertical confined wall for waste heat recovery," *Int. J. Photoenergy* **2022**, 1890541.
- ²⁰D. Haughey and G. Beveridge, "Structural properties of packed beds—a review," *Can. J. Chem. Eng.* **47**, 130–140 (1969).
- ²¹G. F. Al-Sumaily, H. A. Dhahad, and M. C. Thompson, "Mixed convection phenomenon in packed beds: A comprehensive review," *Therm. Sci. Eng. Prog.* **32**, 101242 (2022).
- ²²P. Kumar, S. K. Saha, and A. Sharma, "Experimental and CFD-DEM study on local packing distribution and thermofluidic analysis of binary packed bed," *Chem. Eng. Sci.* **282**, 119372 (2023).
- ²³E. R. Lindgren, "Propagation velocity of turbulent slugs and streaks in transition pipe flow," *Phys. Fluids* **12**, 418–425 (1969).
- ²⁴A. P. Willis and R. R. Kerswell, "Critical behavior in the relaminarization of localized turbulence in pipe flow," *Phys. Rev. Lett.* **98**, 014501 (2007).
- ²⁵H. Wedin and R. R. Kerswell, "Exact coherent structures in pipe flow: Travelling wave solutions," *J. Fluid Mech.* **508**, 333–371 (2004).
- ²⁶A. A. Draad, G. Kuiken, and F. Nieuwstadt, "Laminar-turbulent transition in pipe flow for newtonian and non-Newtonian fluids," *J. Fluid Mech.* **377**, 267–312 (1998).
- ²⁷B. Hof, C. W. Van Doorne, J. Westerweel, F. T. Nieuwstadt, H. Faisst, B. Eckhardt, H. Wedin, R. R. Kerswell, and F. Waleffe, "Experimental observation of nonlinear traveling waves in turbulent pipe flow," *Science* **305**, 1594–1598 (2004).
- ²⁸H. Faisst and B. Eckhardt, "Traveling waves in pipe flow," *Phys. Rev. Lett.* **91**, 224502 (2003).
- ²⁹W. Reichelt, "Zur berechnung des druckverlustes einphasig durchströmter kugel-und zylinderschüttungen," *Chem. Ing. Tech.* **44**, 1068–1071 (1972).
- ³⁰D. Reger, E. Merzari, P. Balestra, S. Schunert, Y. Hassan, and H. Yuan, "Toward development of an improved friction correlation for the near-wall region of pebble bed systems," *arXiv:2203.05041* (2022).
- ³¹D. Pashchenko, R. Mustafin, and A. Mustafina, "Steam methane reforming in a microchannel reformer: Experiment, cfd-modelling and numerical study," *Energy* **237**, 121624 (2021).
- ³²S. Wang, S. A. Nabavi, and P. T. Clough, "A review on bi/polymetallic catalysts for steam methane reforming," *Int. J. Hydrogen Energy* **48**, 15879 (2023).
- ³³S. Saeidi, A. Sápi, A. H. Khoja, S. Najari, M. Ayesha, Z. Kónya, B. B. Asare-Bediako, A. Tatarczuk, V. Hessel, F. J. Keil *et al.*, "Evolution paths from gray to turquoise hydrogen via catalytic steam methane reforming: Current challenges and future developments," *Renewable Sustainable Energy Rev.* **183**, 113392 (2023).
- ³⁴M. Mohandessi, M. Tavakolian, and M. R. Rahimpour, "Cadmium as a robust and novel promoter over Ni@ γ - Al_2O_3 catalysts in dry methane reforming," *ACS Appl. Energy Mater.* **6**, 8209–8220 (2023).
- ³⁵I. Karpilov and D. Pashchenko, "Steam methane reforming over a preheated packed bed: Heat and mass transfer in a transient process," *Therm. Sci. Eng. Prog.* **42**, 101868 (2023).
- ³⁶Y.-L. Lee, K.-J. Kim, G.-R. Hong, and H.-S. Roh, "Target-oriented water-gas shift reactions with customized reaction conditions and catalysts," *Chem. Eng. J.* **458**, 141422 (2023).
- ³⁷S.-Y. Ahn, K.-J. Kim, B.-J. Kim, J.-O. Shim, W.-J. Jang, and H.-S. Roh, "Unravelling the active sites and structure-activity relationship on Cu-ZnO- Al_2O_3 based catalysts for water-gas shift reaction," *Appl. Catal. B* **325**, 122320 (2023).
- ³⁸W.-H. Chen, C.-Y. Lu, W.-S. Chou, A. K. Sharma, A. Saravanakumar, and K.-Q. Tran, "Design and optimization of a crossflow tube reactor system for hydrogen production by combining ethanol steam reforming and water gas shift reaction," *Fuel* **334**, 126628 (2023).
- ³⁹Z. Luo, Y. Lu, D. W. Singer, M. E. Berck, L. A. Somers, B. R. Goldsmith, and A. C. Johnson, "Effect of substrate roughness and feedstock concentration on growth of wafer-scale graphene at atmospheric pressure," *Chem. Mater.* **23**, 1441–1447 (2011).

- ⁴⁰M.-O. Coppens, “The effect of fractal surface roughness on diffusion and reaction in porous catalysts—from fundamentals to practical applications,” *Catal. Today* **53**, 225–243 (1999).
- ⁴¹C. Parra-Cabrera, C. Achille, S. Kuhn, and R. Ameloot, “3D printing in chemical engineering and catalytic technology: Structured catalysts, mixers and reactors,” *Chem. Soc. Rev.* **47**, 209–230 (2018).
- ⁴²G. Imani, L. Zhang, J. Mawaja, H. Sun, D. Fan, M. Ntibanana, L. Hou, Y. Yang, and J. Yao, “Effect of roughness on droplet motion in a capillary channel: A numerical study,” *Phys. Fluids* **35**, 112011 (2023).
- ⁴³D. Kumar and S. Roy, “Simulation of flow over a fixed and oscillating circular cylinder with different surface roughness patterns using a multi-block-multi-mesh framework immersed boundary method,” *Phys. Fluids* **35**, 113606 (2023).
- ⁴⁴J. Chen, Y. Zhang, and S. Fu, “Roughness implementation and convective heat transfer coefficient computation toward ice accretion simulation,” *Phys. Fluids* **35**, 114102 (2023).
- ⁴⁵M. A. Karali, B. A. Almohammadi, A. S. B. Mahfouz, M. A. Abdelmohimen, E.-A. Attia, and H. Refaey, “Effect of surfaces roughness of a staggered tube bank in cross flow with air on heat transfer and pressure drop,” *Case Stud. Therm. Eng.* **43**, 102779 (2023).
- ⁴⁶C. W. Crawford and O. A. Plumb, “The influence of surface roughness on resistance to flow through packed beds,” *J. Fluids Eng.* **108**, 343–347 (1986).
- ⁴⁷R. Jordi, B. Young, and B. V. Vliet, “The effect of surface roughness on pressure drop in a packed bed,” *Chem. Eng. Commun.* **89**, 137–146 (1990).
- ⁴⁸D. Pashchenko, “Experimental investigation of synthesis gas production by methane reforming with flue gas over a Nio-Al₂O₃ catalyst: Reforming characteristics and pressure drop,” *Int. J. Hydrogen Energy* **44**, 7073–7082 (2019).
- ⁴⁹D. Pashchenko, “Experimental investigation of reforming and flow characteristics of a steam methane reformer filled with nickel catalyst of various shapes,” *Energy Convers. Manage.* **185**, 465–472 (2019).
- ⁵⁰V. Papkov, N. Shadymov, and D. Pashchenko, “CFD-modeling of fluid flow in ansys fluent using python-based code for automation of repeating calculations,” *Int. J. Mod. Phys. C* **34**, 2350114 (2023).
- ⁵¹F. R. Menter, “Two-equation eddy-viscosity turbulence models for engineering applications,” *AIAA J.* **32**, 1598–1605 (1994).
- ⁵²Ansyz Fluent, ANSYS FLUENT theory guide, Ansys Inc., USA, 2011.